Massachusetts Institute of Technology

Professional Certificate

COURSE	PREREQUISITES	CONTENT	DURATION
Introduction to Computer Science and Programming	High School – Algebra High School – Reasonable Aptitude for Mathematics Programming Knowledge	1. A Notion of Computation 2. Python Programming Language 3. Some Simple Algorithms 4. Testing and Debugging 5. Informal Intro to Algorithmic Complexity 6. Data Structure	Weeks: 9 Hours: 14 - 16 Cost: 135 USD Instructor-led
Introduction to Computational Thinking and Data Science	6.00.1x / Python Computational Complexity	1. Plotting with the pyLab package 2. Stochastic Programming and Statistical Thinking 3. Monte Carlo Simulations	Weeks: 9 Hours: 14 – 16 Instructor-led
Computational Thinking for Modeling and Simulation	Algebra Calculus	1. What is Computational Thinking 2. Interpolation 3. Integration 4. Randomness 5. Differentiation 6. Solving Equations	Weeks: 9 Hours: 3 – 5 Instructor-led
Micro-Master's Program in Statistics and Data Science		1. Probability – The Science of Uncertainty and Data 2. Fundamentals of Statistics 3. Machine Learning with Python: from Linear Models to Deep Learning 4. Capstone Exam in Statistics and Data Science 5. Data Analysis in Social Science – Assessing Your Knowledge 6. Data Analysis: Statistical Modeling and Computation in Application	1 Year 2 Months Cost: 1350 USD Instructor-led